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Vacuum processing apparatus and oxidizing gas removal method

Abstract

A vacuum processing apparatus includes a decompressable process container; a supply port configured to supply, to the process container, an ionic liquid that absorbs an oxidizing gas; and a discharge port configured to discharge the ionic liquid supplied to the process container. A recess is provided at a joint portion between members constituting the process container. The supply port is configured to supply the ionic liquid to the recess, and the discharge port is configured to discharge the ionic liquid supplied to the recess.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. patent application Ser. No. 17/350,125 filed on Jun. 17, 2021, the entire contents of which are incorporated herein by reference.

BACKGROUND
1. Technical Field
(1) The present disclosure relates to a vacuum processing apparatus and an oxidizing gas removal method.
2. Background Art
(2) A technique is known in which a graphene structure is formed on the surface of a substrate housed in a process container by a remote microwave plasma CVD using a carbon-containing gas as a deposition raw material gas (see, for example, Patent Document 1).
(3) The present disclosure provides a technique that enables the removal of an oxidizing gas remaining in a process container.
PRIOR ART DOCUMENT
Patent Document
(4) [Patent Document 1] Japanese Laid-open Patent Publication No. 2019-55887
SUMMARY OF THE INVENTION
(5) According to one aspect of the present disclosure, a vacuum processing apparatus includes: a decompressable process container; a supply port that is formed on a side wall of the process container and that is configured to supply, to the process container, an ionic liquid that absorbs an oxidizing gas; and a discharge port configured to discharge the ionic liquid supplied to the process container.
(6) According to another aspect of the present disclosure, a vacuum processing apparatus includes a decompressable process container; a supply port configured to supply, to the process container, an ionic liquid that absorbs an oxidizing gas; and a discharge port configured to discharge the ionic liquid supplied to the process container. A recess is provided at a joint portion between members constituting the process container. The supply port is configured to supply the ionic liquid to the recess, and the discharge port is configured to discharge the ionic liquid supplied to the recess.
(7) According to the present disclosure, it is possible to remove an oxidizing gas remaining in a process container.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a schematic diagram illustrating an example of a vacuum processing apparatus according to a first embodiment;
(2) FIG. **2** is a schematic diagram illustrating an example of a vacuum processing apparatus according to a second embodiment;
(3) FIG. **3** is a schematic diagram illustrating an example of a vacuum processing apparatus according to a third embodiment;
(4) FIG. **4** is a schematic diagram illustrating an example of a vacuum processing apparatus according to a fourth embodiment; and
(5) FIG. **5** is an enlarged view of a joint portion between a process container and a support member in the vacuum processing apparatus of FIG. **4**.
DESCRIPTION OF THE EMBODIMENTS
(6) Hereinafter, non-limiting exemplary embodiments of the present disclosure will be described

with reference to the accompanying drawings. In all the accompanying drawings, the same or corresponding reference numerals shall be attached to the same or corresponding components and overlapping descriptions may be omitted.
(7) [Oxidizing Gas Remaining in Process Container]
(8) In a state in which an oxidizing gas such as H.sub.2O gas or O.sub.2 gas is adsorbed on the inner wall of a process container, when a substrate is housed in the process container and a semiconductor process such as deposition or and etching is implemented, the oxidizing gas may be desorbed from the inner wall of the process container and may adversely affect the semiconductor process.
(9) For example, as an example of a semiconductor process, in a case in which a graphene film is formed on the surface of a wiring layer formed on the surface of a substrate, an oxidizing gas desorbed from the inner wall of a process container oxidizes the surface of the wiring layer of the substrate before forming the graphene film, and an oxide film is formed at the interface between the wiring layer and the graphene film. Since the oxide formed at the interface between the wiring layer and the graphene film has insulating properties, the contact resistance between the wiring layer and the graphene film is increased.
(10) Accordingly, the present disclosure provides a technique of supplying an ionic liquid to a process container and causing an oxidizing gas remaining in the process container to be absorbed by the ionic liquid, thereby removing the oxidizing gas remaining in the process container. The details will be described below.
(11) <Vacuum Processing Apparatus>
First Embodiment
(12) Referring to FIG. **1**, an example of a vacuum processing apparatus **1**A according to a first embodiment will be described. The vacuum processing apparatus **1**A illustrated in FIG. **1** may be configured, for example, as a plasma processing apparatus of a RLSA (Radial Line Slot Antenna) microwave plasma system.
(13) The vacuum processing apparatus **1**A includes an apparatus body **10** and a controller **11** that controls the apparatus body **10**.
(14) The apparatus body **10** includes a chamber **101**, a stage **102**, a microwave introduction mechanism **103**, a gas supply mechanism **104**, an exhaust mechanism **105**, a liquid supply mechanism **106**, and the like.
(15) The chamber **101** is generally cylindrically formed. An opening **110** is formed in the substantially central portion of a bottom wall **101***a* of the chamber **101**. The bottom wall **101***a* is provided with an exhaust chamber **111** that is in communication with the opening **110** and that protrudes downward. On a side wall **101***s* of the chamber **101**, a transportation port **117** is formed through which the substrate W passes. The transportation port **117** is opened and closed by a gate valve **118**. The chamber **101**, together with a portion of the microwave introduction mechanism **103**, constitutes a process container of which the inside is decompressable.
(16) A substrate W to be processed is mounted on the stage **102**. The stage **102** has a generally disk shape. The stage **102** is made of ceramics such as aluminum nitride (AlN). The stage **102** is supported by a support post **112** that has a generally cylindrical shape extending upward from the substantially center of the bottom of the exhaust chamber **111** and is made of ceramics such as AlN. An edge ring **113** is provided at the outer edge of the stage **102** so as to surround the substrate W mounted on the stage **102**. Inside the stage **102**, a lifting/lowering pin (not illustrated) for lifting and lowering the substrate W is provided to be able to protrude/retract with respect to the upper surface of the stage **102**.
(17) Inside the stage **102**, a resistance heating type heater **114** is embedded. The heater **114** heats the substrate W mounted on the stage **102** in response to power supplied from a heater power source **115**. Also, a thermocouple (not illustrated) is inserted in the stage **102** to control the temperature of the substrate W to, for example, 350° C. to 850° C., based on a signal from the

thermocouple. In addition, within the stage **102**, an electrode **116** having the same size as the substrate W is embedded above the heater **114**. A bias power source **119** is electrically connected to the electrode **116**. The bias power source **119** supplies predetermined electric power having a predetermined frequency and magnitude to the electrode **116**. This causes ions to be attracted on the substrate W mounted on the stage **102**. It should be noted that the bias power source **119** may not be provided depending on the characteristics of a plasma process.

(18) The microwave introduction mechanism **103** is provided on the upper section of the chamber **101**. The microwave introduction mechanism **103** includes an antenna **121**, a microwave output section **122**, a microwave transmission mechanism **123**, and the like. The antenna **121** is formed with a number of slots **121***a* that are through holes. The microwave output section **122** outputs a microwave. The microwave transmission mechanism **123** directs the microwave output from the microwave output section **122** to the antenna **121**.

(19) Below the antenna **121**, a dielectric window **124** made of a dielectric material is provided. The dielectric window **124** is supported on a support member **132**, which is annularly arranged on the upper section of the chamber **101**. A target **140** made of metal is arranged on the lower surface (surface facing the stage **102**) of the dielectric window **124**. The target **140** includes, for example, at least one metal of titanium, cobalt, aluminum, yttrium, aluminum nitride, and titanium nitride. A shielding member **125** and a wave delay plate **126** are provided on the antenna **121**. A refrigerant flow path (not illustrated) is provided within the shielding member **125**. The shielding member **125** cools the antenna **121**, the dielectric window **124**, the wave delay plate **126**, and the target **140** by a cooling fluid such as water flowing in the refrigerant flow path.

(20) The antenna **121** may be formed, for example, of an aluminum plate or a copper plate having a silver or gold-plated surface. A plurality of slots **121***a* for emitting a microwave are arranged in the antenna **121** in a predetermined pattern. The arrangement pattern of the slots **121***a* is suitably set so that the microwave is emitted evenly. An example of a suitable pattern may be radial line slots in which a plurality of pairs of slots **121***a* including two T-shaped arranged slots **121***a* as one pair are arranged concentrically. The length and array interval of the slots **121***a* are suitably set in accordance with the effective wavelength of the microwave (λg). The slots **121***a* may also have other shapes, such as a circular shape or an arc shape. Further, the arrangement configuration of the slots **121***a* is not particularly limited, and may be an arrangement of helical or radial other than concentric, for example.

(21) The pattern of slots **121***a* is suitably set to be microwave emission characteristics that enable to obtain a desired plasma density distribution.

(22) The wave delay plate **126** is made of a dielectric material having a dielectric constant greater than vacuum such as quartz, ceramics (Al.sub.2O.sub.3), polytetrafluoroethylene, or polyimide. The wave delay plate **126** has a function to make the antenna **121** smaller by shortening the wavelength of the microwave to shorter than in vacuum. It should be noted that the dielectric window **124** is made of a similar dielectric material.

(23) The thickness of the dielectric window **124** and the wave delay plate **126** is adjusted so that an equivalent circuit formed by the wave delay plate **126**, the antenna **121**, the dielectric windows **124**, the target **140**, and the plasma satisfy the resonance conditions. By adjusting the thickness of the wave delay plate **126**, the phase of the microwave can be adjusted. By adjusting the thickness of the wave delay plate **126** so that the joint portion of the antenna **121** is an anti-node of a standing wave, the reflection of the microwave can be minimized and the emission energy of the microwave can be maximized. In addition, by making the wave delay plate **126** and the dielectric window **124** with the same material, it is possible to prevent interfacial reflection of the microwave.

(24) The microwave output section **122** has a microwave oscillator. The microwave oscillator may be of a magnetron type or of a solid state type. The frequency of the microwave that is generated by the microwave oscillator may be, for example, a frequency of 300 MHz to 10 GHz. As an example, the microwave output section **122** outputs a microwave of 2.45 GHz by a magnetron type

microwave oscillator. The microwave is an example of an electromagnetic wave.
(25) The microwave transmission mechanism **123** includes a waveguide **127**, a coaxial waveguide **128**, a mode conversion mechanism **131**, and the like. The waveguide **127** directs a microwave output from the microwave output section **122**. The coaxial waveguide **128** includes an inner conductor **129** connected to the center of the antenna **121** and an outer conductor **130**. The mode conversion mechanism **131** is provided between the waveguide **127** and the coaxial waveguide **128**. The microwave output from the microwave output section **122** propagates in the waveguide **127** in the TE mode and is converted from the TE mode to the TEM mode by the mode conversion mechanism **131**. The microwave converted to the TEM mode propagates through the coaxial waveguide **128** to the wave delay plate **126** and is emitted from the wave delay plate **126** into the chamber **101** via the slots **121***a* of the antenna **121**, the dielectric window **124**, and the target **140**. It should be noted that a tuner (not illustrated) is provided in the middle of the waveguide **127** so that the impedance of the load (plasma) in the chamber **101** matches the output impedance of the microwave output section **122**.
(26) The gas supply mechanism **104** includes a shower ring **142**. The shower ring **142** is annularly arranged along the inner wall of the chamber **101**. The shower ring **142** includes an annular flow path **166** provided therein and a plurality of ejection ports **167** connected to the flow path **166** and opened therein. A gas supply section **163** is connected to the flow path **166** via a pipe **161**. The gas supply section **163** includes a plurality of gas sources, a plurality of flow controllers, and the like. The gas supply section **163** is configured to supply at least one process gas from a corresponding gas source via a corresponding flow controller to the shower ring **142**. The gas supplied to the shower ring **142** is supplied into the chamber **101** from the plurality of ejection ports **167**.
(27) In a case which a metal film is deposited on the substrate W, the gas supply section **163** supplies an inert gas controlled to a predetermined flow rate into the chamber **101** via the shower ring **142**. The inert gas may be, for example, a noble gas or a nitrogen (N.sub.2) gas. Alternatively, in a case in which a metal film is formed on the substrate W, the gas supply section **163** may supply, in addition to the inert gas, a reducing gas into the chamber **101** via the shower ring **142**. The reducing gas may be, for example, a hydrogen-containing gas or a halogen-containing gas.
(28) Also, in a case in which a graphene film is deposited on the substrate W, the gas supply section **163** supplies a carbon-containing gas, a hydrogen-containing gas, and a noble gas controlled to predetermined flow rates into the chamber **101** via the shower ring **142**. The carbon-containing gas may be, for example, a C.sub.2H.sub.2 gas, a C.sub.2H.sub.4 gas, a CH.sub.4 gas, a C.sub.2H.sub.6 gas, a C.sub.3H.sub.8 gas, a C.sub.3H.sub.6 gas, or a combination thereof. The hydrogen-containing gas may be, for example, a hydrogen (H.sub.2) gas. It should be noted that a halogen-based gas such as a fluorine (F.sub.2) gas, a chlorine (Cl.sub.2) gas, or a bromine (Br.sub.2) gas may be used instead of or in addition to the H.sub.2 gas. The noble gas may be, for example, an argon (Ar) gas or a helium (He) gas.
(29) An exhaust mechanism **105** includes an exhaust chamber **111**, an exhaust pipe **181**, an exhaust device **182**, and the like. The exhaust pipe **181** is provided on the side wall of the exhaust chamber **111**. The exhaust device **182** is connected to the exhaust pipe **181**. The exhaust device **182** includes a vacuum pump, a pressure control valve, and the like.
(30) The liquid supply mechanism **106** supplies an ionic liquid into the chamber **101** and collects the ionic liquid supplied in the chamber **101**. The ionic liquid is an ionic liquid that absorbs an oxidizing gas. Details of the ionic liquid that absorbs an oxidizing gas will be described later. The liquid supply mechanism **106** includes a supply port **191**, a discharge port **192**, a liquid circulating section **193**, and the like.
(31) The supply port **191** is formed through the side wall **101***s* of the chamber **101**. The supply port **191** supplies the ionic liquid from the side of the chamber **101** into the chamber **101**. The ionic liquid supplied into the chamber **101** flows downward along the inner wall of the chamber **101**, as indicated by the arrow A**1** in FIG. **1**. FIG. **1** illustrates a case where the supply port **191** is formed

below the transportation port **117**. However, the supply port **191** may be formed above the transportation port **117**.
(32) The discharge port **192** is formed so as to penetrate the bottom of the exhaust chamber **111**. The discharge port **192** discharges the ionic liquid supplied in the chamber **101** to the outside of the chamber **101**. In FIG. **1**, a case is illustrated in which the discharge port **192** is formed at the bottom of the exhaust chamber **111**. It should be noted that the discharge port **192** may be formed on the side wall of the exhaust chamber **111** or on the bottom wall **101***a* of the chamber **101**. Also, a plurality of discharge ports **192** may be formed.
(33) The liquid circulating section **193** collects the ionic liquid discharged from the discharge port **192** and introduces the collected ionic liquid into the supply port **191** to circulate the ionic liquid. The liquid circulating section **193** includes a tank **193***a*, a temperature control mechanism **193***b*, a forward pipe **193***c*, a return pipe **193***d*, and the like.
(34) The tank **193***a* is connected to the discharge port **192** via the return pipe **193***d*. The tank **193***a* stores the ionic liquid that is discharged from the discharge port **192**.
(35) Temperature control mechanism **193***b* includes a heater, a temperature sensor (neither illustrated), and the like. The temperature control mechanism **193***b* controls the temperature of the ionic liquid in the tank **193***a* by controlling the heater based on the detected value of the temperature sensor.
(36) The forward pipe **193***c* connects the supply port **191** to the tank **193***a*. The forward pipe **193***c* introduces the ionic liquid stored in the tank **193***a* into the supply port **191**. A valve **193***e* is interposed on the forward pipe **193***c*. When the valve **193***e* is opened, the ionic liquid is introduced from within the tank **193***a* into the supply port **191**, and when the valve **193***e* is closed, the introduction of the ionic liquid from within the tank **193***a* into the supply port **191** is stopped.
(37) The return pipe **193***d* connects the discharge port **192** to the tank **193***a*. The return pipe **193***d* collects the ionic liquid discharged from the discharge port **192** into the tank **193***a*. A valve **193***f* is interposed on the return pipe **193***d*. When the valve **193***f* is opened, the ionic liquid is collected from the discharge port **192** into the tank **193***a*, and when the valve **193***f* is closed, the collection of the ionic liquid from the discharge port **192** into the tank **193***a* is stopped.
(38) The controller **11** includes a memory, a processor, an input/output interface, and the like. The memory contains a recipe including a program executed by the processor and the conditions of each process. The processor executes a program read from the memory and controls each section of the apparatus body **10** through the input/output interface based on the recipe stored in the memory.
(39) For example, the controller **11** performs a method of removing oxidizing gas prior to carrying out a semiconductor process in a process container housing a substrate. Specifically, the controller **11** controls to open the valves 193*e* and 193*f* prior to the semiconductor process. This causes the ionic liquid to be supplied from the supply port **191** into the chamber **101**, and the supplied ionic liquid absorbs the oxidizing gas remaining in the chamber **101**. The ionic liquid absorbing the oxidizing gas is discharged out of the chamber **101** through the discharge port **192**. It should be noted that the controller **11** may perform the method of removing the oxidizing gas at a different timing from that before performing the semiconductor process.
(40) As described above, the vacuum processing apparatus **1**A according to the first embodiment includes the supply port **191** formed on the side wall **101***s* of the chamber **101** to supply the ionic liquid into the chamber **101** and the discharge port **192** for discharging the ionic liquid supplied in the chamber **101**. This allows the ionic liquid to be supplied from the supply port **191** into the chamber **101** to absorb the oxidizing gas remaining in the chamber **101** by the ionic liquid. In addition, the ionic liquid absorbing the oxidizing gas can be discharged from the discharge port **192** to the outside of the chamber **101**. As a result, the oxidizing gas remaining in the chamber **101** can be removed.
(41) It should be noted that although a case has been described in the first embodiment in which one supply port **191** is formed, the present disclosure is not limited to this. For example, a plurality

of supply ports **191** may be formed. In a case in which a plurality of supply ports **191** are formed, the plurality of supply ports **191** are preferably formed at intervals along the circumferential direction of the side walls **101***s* of the chamber **101**. This enables the ionic liquid to be ejected from the plurality of positions in the circumferential direction of the chamber **101**. Therefore, the ionic liquid flows over a wide range of inner wall of the chamber **101**, and the surface area of the ionic liquid flowing in the chamber **101** becomes large. As a result, the absorption efficiency of the oxidizing gas remaining in the chamber **101** is increased.

Second Embodiment

(42) Referring to FIG. **2**, an example of a vacuum processing apparatus **1**B according to a second embodiment will be described. The vacuum processing apparatus **1**B according to the second embodiment differs from the vacuum processing apparatus **1**A according to the first embodiment in that a groove **101***g* is formed on the inner wall of the chamber **101** to flow the ionic liquid along the circumferential direction of the inner wall. Hereinafter, different points from the vacuum processing apparatus **1**A will be mainly described.

(43) The groove **101***g* is spirally formed along the circumferential direction of the inner wall of the chamber **101**. The upper side end of the groove **101***g* communicates with the supply port **191** to allow the ionic liquid supplied from the supply port **191** to flow along the circumferential direction of the inner wall of the chamber **101**. Therefore, because the ionic liquid flows over a wide range of inner wall of the chamber **101**, the surface area of the ionic liquid flowing in the chamber **101** becomes large. As a result, the absorption efficiency of the oxidizing gas remaining in the chamber **101** is increased.

(44) As described above, the vacuum processing apparatus **1**B according to the second embodiment includes the supply port **191** formed in the side wall **101***s* of the chamber **101** to supply the ionic liquid into the chamber **101** and the discharge port **192** for discharging the ionic liquid supplied into the chamber **101**. This allows the ionic liquid to be supplied from the supply port **191** into the chamber **101** to absorb the oxidizing gas remaining in the chamber **101** by the ionic liquid. In addition, the ionic liquid absorbing the oxidizing gas can be discharged from the discharge port **192** to the outside of the chamber **101**. As a result, the oxidizing gas remaining in the chamber **101** can be removed.

(45) Further, according to the vacuum processing apparatus **1**B according to the second embodiment, the groove **101***g* for flowing the ionic liquid along the circumferential direction of the inner wall is formed on the inner wall of the chamber **101**. This causes the ionic liquid to flow along the groove **101***g* within the chamber **101**. Therefore, the ionic liquid flows over a wide range of inner wall of the chamber **101**, and the surface area of the ionic liquid flowing in the chamber **101** becomes large. Also, because the path from the supply port **191** to the discharge port **192** becomes long, the time for the ionic liquid to flow in the chamber **101** becomes long. As a result, the absorption efficiency of the oxidizing gas remaining in the chamber **101** is increased.

(46) It should be noted that although a case has been described in the second embodiment in which one supply port **191** is formed, the present disclosure is not limited to this. For example, a plurality of supply ports **191** may be formed. In a case in which a plurality of supply ports **191** are formed, it is preferable that the plurality of supply ports **191** are formed at intervals along the circumferential direction of the side wall **101***s* of the chamber **101**, and that a groove **101***g* is formed corresponding to each of the plurality of supply ports **191**. Therefore, because the ionic liquid flows over a wide range of inner wall of the chamber **101**, the surface area of the ionic liquid flowing in the chamber **101** becomes large. As a result, the absorption efficiency of the oxidizing gas remaining in the chamber **101** is increased.

Third Embodiment

(47) Referring to FIG. **3**, an example of a vacuum processing apparatus **1**C according to a third embodiment will be described. The vacuum processing apparatus **1**C according to the third embodiment differs from the vacuum processing apparatus **1**A of the first embodiment in including

a liquid supply mechanism **306** including a supply port **391** provided annularly along the inner wall of the chamber **101** instead of the supply port **191**. Hereinafter, different points from the vacuum processing apparatus **1**A will be mainly described.
(48) The liquid supply mechanism **306** supplies an ionic liquid into the chamber **101** and collects the ionic liquid supplied into the chamber **101**. The ionic liquid is an ionic liquid that absorbs an oxidizing gas. The liquid supply mechanism **306** includes the supply port **391**, the discharge port **192**, the liquid circulating section **193**, and the like.
(49) The supply port **391** is provided annularly along the inner wall of the chamber **101**. The supply port **391** includes an annular liquid flow path **391***a* provided inside and a plurality of liquid ejection ports **391***b* connected to and opened inside the liquid flow path **391***a*. The tank **193***a* is connected to the liquid flow path **391***a* through the forward pipe **193***c*. While flowing in the circumferential direction of the chamber **101** along the liquid flow path **391***a*, the ionic liquid introduced to the supply port **391** is supplied from the plurality of liquid ejection ports **391***b* into the chamber **101** as indicated by the arrow A**3** in FIG. **3** and flows downward along the inner wall of the chamber **101**. Therefore, because the ionic liquid flows over a wide range of inner wall of the chamber **101**, the surface area of the ionic liquid flowing in the chamber **101** becomes large. As a result, the absorption efficiency of the oxidizing gas remaining in the chamber **101** is increased.
(50) As described above, the vacuum processing apparatus **1**C according to the third embodiment includes the supply port **391** formed on the side wall **101***s* of the chamber **101** to supply the ionic liquid into the chamber **101** and the discharge port **192** for discharging the ionic liquid supplied in the chamber **101**. This allows the ionic liquid to be supplied from the supply port **391** into the chamber **101** to absorb the oxidizing gas remaining in the chamber **101** by the ionic liquid. In addition, the ionic liquid absorbing the oxidizing gas can be discharged from the discharge port **192** to the outside of the chamber **101**. As a result, the oxidizing gas remaining in the chamber **101** can be removed.
(51) Also, according to the vacuum processing apparatus **1**C of the third embodiment, the supply port **391** includes the annular liquid flow path **391***a* provided inside and the plurality of liquid ejection ports **391***b* connected to and opened inside the liquid flow path **391***a*. As a result, the ionic liquid can be ejected from the plurality of positions in the circumferential direction of the chamber **101**. Therefore, the ionic liquid flows over a wide range of inner wall of the chamber **101**, and the surface area of the ionic liquid flowing in the chamber **101** becomes large. As a result, the absorption efficiency of the oxidizing gas remaining in the chamber **101** is increased.
(52) It should be noted that although, in the third embodiment described above, the vacuum processing apparatus **1**C includes the liquid supply mechanism **306** including the supply port **391** provided in an annular shape along the inner wall of the chamber **101** in lieu of the supply port **191** included in the vacuum processing apparatus **1**A, the present disclosure is not limited to this. For example, the vacuum processing apparatus **1**C may include both a supply port **191** and a supply port **391**.
Fourth Embodiment
(53) An example of a vacuum processing apparatus **1**D according to a fourth embodiment will be described with reference to FIG. **4** and FIG. **5**. The vacuum processing apparatus **1**D according to the fourth embodiment differs from the vacuum processing apparatus **1**A according to the first embodiment in that a recess **101***h* for flowing an ionic liquid is formed at a joint portion between the members constituting the process container. Hereinafter, different points from the vacuum processing apparatus **1**A will be mainly described.
(54) A sealing material **151** is provided at the joint portion of the chamber **101** and a support member **132** to seal the joint portion. The sealing material **151** may be, for example, an O-ring.
(55) The recess **101***h* is annularly formed along the sealing material **151** on the vacuum side of the sealing material **151** at the joint portion of the chamber **101** and the support member **132**. The recess **101***h* is in communication with the supply port **491**, and the ionic liquid is supplied to the

recess **101***h* through the supply port **491**. The ionic liquid supplied to the recess **101***h* flows along the circumferential direction of the chamber **101** along the recess **101***h*.

(56) The recess **101***h* is preferably formed so that the ionic liquid flowing through the recess **101***h* contacts both the chamber **101** and the support member **132**. This allows the ionic liquid flowing through the recess **101***h* to function as an annular conductive sealing material represented by a spiral seal. That is, the ionic liquid flowing through the recess **101***h* ensures continuity between the chamber **101** and the support member **132** and keeps the support member **132** at the ground potential. The ionic liquid flowing through the recess **101***h* also prevents the leakage of high frequency or plasma from between the chamber **101** and the support member **132**. It should be noted that in a case in which the ionic liquid flowing in the recess **101***h* is used instead of a spiral seal, the ionic liquid flowing in the recess **101***h* is recovered by opening the valve **193***f* when the chamber **101** is opened to atmosphere by maintenance or the like. Further, when the maintenance or the like is completed and the inside of the chamber **101** is depressurized, the valve **193***f* is closed and the valve **193***e* is opened to fill the recess **101***h* with the ionic liquid and then the inside of the chamber **101** is depressurized. Thus, since the ionic liquid filled in the recess **101***h* absorbs the oxidizing gas within the chamber **101**, the chamber **101** can be allowed to be in a high vacuum and an environment with less oxidizing gas can be formed.

(57) The recess **101***h* has a depth on the vacuum side deeper than the depth on the sealing material **151** side, as illustrated in FIG. **5**. Thus, the ionic liquid filled in the recess **101***h* is suppressed from flowing out to the vacuum side, and therefore the state in which the recess **101***h* is filled with the ionic liquid can be maintained.

(58) The supply port **491** is formed so as to penetrate the support member **132**. The supply port **491** is in communication with the recess **101***h* and supplies the ionic liquid from the side of the process container to the recess **101***h* in the process container.

(59) The discharge port **492** is formed to penetrate the side wall **101***s* of the chamber **101** and is in communication with the recess **101***h*. The discharge port **492** discharges the ionic liquid flowing through the recess **101***h* to the outside of the chamber **101**.

(60) As described above, the vacuum processing apparatus **1**D according to the fourth embodiment includes the supply port **491** formed on the side wall (support member **132**) of the process container to supply the ionic liquid to the recess **101***h* and the discharge port **492** that discharges the ionic liquid supplied to the recess **101***h*. This allows the ionic liquid to be supplied from the supply port **491** to the recess **101***h* to absorb the oxidizing gas remaining in the chamber **101** by the ionic liquid. In addition, the ionic liquid absorbing the oxidizing gas can be discharged from the discharge port **492** to the chamber **101**. As a result, the oxidizing gas remaining in the chamber **101** can be removed.

(61) In addition, according to the vacuum processing apparatus **1**D of the fourth embodiment, the ionic liquid flowing through the recess **101***h* contacts both the chamber **101** and the support member **132**. This allows the ionic liquid flowing through the recess **101***h* to function as an annular conductive sealing material instead of a spiral seal. That is, the ionic liquid flowing through the recess **101***h* ensures continuity between the chamber **101** and the support member **132** and keeps the support member **132** at the ground potential. The ionic liquid flowing through the recess **101***h* also prevents the leakage of high frequency or plasma from between the chamber **101** and the support member **132**.

(62) It should be noted that although the depth on the vacuum side is deeper than the depth on the sealing material **151** side of the recess **101***h* in the described fourth embodiment, the present disclosure is not limited thereto. For example, the depth of the vacuum side and the depth of the sealing material **151** side of the recess **101***h* may be the same. In this case, a portion of the ionic liquid flowing through the recess **101***h* flows into the inner wall of the chamber **101**, thereby increasing the surface area of the ionic liquid flowing through the chamber **101**. As a result, the absorption efficiency of the oxidizing gas remaining in the chamber **101** is increased. In this case,

in addition to the discharge port **492** communicating with the recess **101***h*, by providing a discharge port formed to penetrate the bottom of the exhaust chamber **111**, the ionic liquid flowing into the inner wall of the chamber **101** can be discharged to the outside of the chamber **101**.
(63) Although, in the fourth embodiment described above, the vacuum processing apparatus **1**D includes the liquid supply mechanism **406** including the supply port **491** formed to penetrate the support member **132** instead of the supply port **191** included in the vacuum processing apparatus **1**A, the present disclosure is not limited to this. For example, the vacuum processing apparatus **1**D may include both the supply port **191** and the supply port **491**. For example, the vacuum processing apparatus **1**D may include the groove **101***g* included in the vacuum processing apparatus **1**B and may include the supply port **391** included in the vacuum processing apparatus **1**C.
(64) Ionic Liquid
(65) In the vacuum processing apparatuses **1**A to **1**D, an example of a suitably available ionic liquid will be described. The ionic liquid is an ionic liquid that absorbs an oxidizing gas. Examples of the oxidizing gas include H.sub.2O gas and O.sub.2 gas.
(66) In a case in which the oxidizing gas to be absorbed by the ionic liquid is H.sub.2O gas, the ionic liquid having a molecular structure with a high polarity can be suitably used as the ionic liquid. By using the ionic liquid having a molecular structure with a high polarity, the ionic liquid can efficiently absorb H.sub.2O, which is a polar molecule. Examples of such an ionic liquid include DEME-BF.sub.4 (N,N-Diethyl-N-methyl-N-(2-methoxyethyl)ammonium tetrafluoroborate) and EMI-AcO (1-Ethyl-3-methylimidazolium-acetate). Also, it may be a halide-based ionic liquid such as [BPy]Cl represented by chemical formula I1, [B2MPY]Cl represented by chemical formula I2, [B3MPy]Cl represented by chemical formula I3, and [B4MPY]Cl represented by chemical formula I4.
(67) ##STR00001##
(68) In a case in which the oxidizing gas to be absorbed by the ionic liquid is H.sub.2O gas and O.sub.2 gas, the ionic liquid having a molecular structure including a non-polar portion can be preferably used as the ionic liquid, for example. In general, because many ionic liquids, which are combinations of anions and cations, have polarity, it is considered that nonpolar O.sub.2 molecules are not easily absorbed by an ionic liquid with polarity. Therefore, by using ionic liquid having a molecular structure including a non-polar portion, O.sub.2 gas can be absorbed efficiently by the nonpolar portion contained in the ionic liquid. Such an ionic liquid may be MEMP (N-(2-methoxyethyl)-N-methyl-pyrrolidinium)-TFSI (bis(tri-fluoro-methane-sulfonyl)imide).
(69) The embodiments disclosed herein should be considered exemplary in all respects and are not limited thereto. The embodiments as described above may be, omitted, substituted, and changed in various forms without departing from the appended claims and spirit thereof.
(70) In the above embodiments, although a case has been described in which a liquid circulating section for introducing and circulating an ionic liquid discharged from a discharge port to a supply port, the present disclosure is not limited thereto. For example, an ionic liquid may be introduced from an ionic liquid supply source into a supply port without providing a liquid circulation section.
(71) In the embodiments described above, the vacuum processing apparatus is configured as a cold wall type apparatus, but the present disclosure is not limited thereto. The ionic liquid does not volatilize even in vacuum and has heat resistance. Thus, the vacuum processing apparatus may be a hot wall type apparatus in which the wall surface of the process container is heated to a high temperature.
(72) In the embodiments described above, the vacuum processing apparatus is configured as a plasma processing apparatus, but the present disclosure is not limited thereto. The vacuum processing apparatus is not limited to a plasma processing apparatus as long as it is an apparatus that applies a predetermined process (e.g., deposition, etching) to a substrate. For example, the vacuum processing apparatus may be an ALD (Atomic Layer Deposition) apparatus, a CVD (Chemical Vapor Deposition) apparatus, a PVD (Physical Vapor Deposition) apparatus, or the like.

Claims

1. A vacuum processing apparatus comprising: a decompressable process container; the process container including a support member and a side wall of the process container, wherein the support member is provided on the side wall of the process container and is configured to support a dielectric window; a tank storing an ionic liquid that absorbs an oxidizing gas; a supply port connected to the tank and configured to supply the ionic liquid from the tank to the process container; and a discharge port configured to discharge the ionic liquid supplied to the process container, wherein a recess is annularly provided at a joint portion between the support member and the side wall of the process container, wherein the supply port is configured to supply the ionic liquid to the recess, and the discharge port is configured to discharge the ionic liquid supplied to the recess, wherein the vacuum processing apparatus further comprises a sealing material configured to seal the joint portion from an outside of the process container, wherein the recess is provided on a vacuum side of the sealing material at the joint portion and that is in communication with the supply port and an inside of the process container, and the ionic liquid supplied from the supply port flows through the recess along the sealing material, wherein the discharge port is provided in communication with the recess, the discharge port being located at a bottom of the recess, wherein a top surface of an inner annular rim of the side wall of the process container is higher than a bottom surface of the support member, and wherein the ionic liquid flowing through the recess is in contact with the sealing material and the bottom surface of the support member, but not the top surface of the inner annular rim of the side wall of the process container.
2. The vacuum processing apparatus according to claim 1, wherein the ionic liquid is MEMP (N-(2-methoxyethyl)-N-methyl-pyrrolidinium)-TFSI(bis(tri-fluoro-methane-sulfonyl)imide).
3. A method for removing an oxidizing gas in a vacuum processing apparatus including: a decompressable process container; the process container including a support member and a side wall of the process container, wherein the support member is provided on the side wall of the process container and is configured to support a dielectric window; a tank storing an ionic liquid that absorbs an oxidizing gas; a supply port connected to the tank and configured to supply the ionic liquid from the tank to the process container; and a discharge port configured to discharge the ionic liquid supplied to the process container, wherein a recess is annularly provided at a joint portion between the support member and the side wall of the process container, wherein the supply port is configured to supply the ionic liquid to the recess, and the discharge port is configured to discharge the ionic liquid supplied to the recess, wherein the vacuum processing apparatus further comprises a sealing material configured to seal the joint portion from an outside of the process container, wherein the recess is provided on a vacuum side of the sealing material at the joint portion and that is in communication with the supply port and an inside of the process container, and the ionic liquid supplied from the supply port flows through the recess along the sealing material, wherein the discharge port is provided in communication with the recess, the discharge port being located at a bottom of the recess, wherein a top surface of an inner annular rim of the side wall of the process container is higher than a bottom surface of the support member, and wherein the ionic liquid flowing through the recess is in contact with the sealing material and the bottom surface of the support member, but not the top surface of the inner annular rim of the side wall of the process container, the method comprising: supplying the ionic liquid that absorbs the oxidizing gas to the recess from the supply port, thereby absorbing the oxidizing gas remaining in the process container by the ionic liquid; and discharging the ionic liquid supplied to the recess.